DSA MARATHON

Day 3 (Wednesday) — 20 Coding Questions

With Complete C++ Programs

Question 1: Sum and Average of Three Numbers

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b,c;
  cin>>a>>b>>c;
  int sum=a+b+c;
  double avg=static_cast<double>(sum)/3;
  cout<<"Sum = "<<sum<<endl;
  cout<<"Average = "<<avg<<endl;
  return 0;
}
```

Question 2: Perform Multiple Arithmetic Operations

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b;
  cin>>a>>b;
  cout<<"Addition = "<<a+b<<endl;
  cout<<"Subtraction = "<<a-b<<endl;
  cout<<"Multiplication = "<<a*b<<endl;
  cout<<"Division = "<<a/b<<endl;
  cout<<"Remainder = "<<a%b<<endl;
  return 0;
}
```

Question 3: Arithmetic Operations on Four Numbers

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b,c,d;
  cin>>a>>b>>c>>d;
  int sum=a+b+c+d;
  int product=a*b*c*d;
  double avg=static_cast<double>(sum)/4;
```

```
  cout<<sum<<endl;
  cout<<product<<endl;
  cout<<avg<<endl;
  return 0;
}
```

Question 4: Area and Perimeter of Rectangle

```
#include <iostream>
#include <string>
using namespace std;
int main(){
  int l,b;
  cin>>l>>b;
  cout<<"Area = "<<l*b<<endl;
  cout<<"Perimeter = "<<2*(l+b)<<endl;
  return 0;
}
```

Question 5: Area and Circumference of Circle

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  double r;
  cin>>r;
  cout<<"Area = "<<3.14159*r*r<<endl;
  cout<<"Circumference = "<<2*3.14159*r<<endl;
  return 0;
}
```

Question 6: Student Percentage Calculator

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b,c,d,e;
  cin>>a>>b>>c>>d>>e;
  int total=a+b+c+d+e;
  double per=static_cast<double>(total)/5;
  cout<<total<<endl;
  cout<<per<<endl;
  return 0;
}
```

Question 7: Swap Two Numbers (Using Third Variable)

```
#include <iostream>
#include <string>
using namespace std;
```

```
int main()
{
  int a,b,temp;
  cin>>a>>b;
  temp=a;
  a=b;
  b=temp;
  cout<<a<<" "<<b<<endl;
  return 0;
}
```

Question 8: Swap Two Numbers (Without Third Variable)

```
#include <iostream>
#include <string>
using namespace std;
int main()
{ int a,b;
  cin>>a>>b;
  a=a+b;
  b=a-b;
  a=a-b;
  cout<<a<<" "<<b<<endl;
  return 0;
}
```

Question 9: ASCII Value Finder

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  char ch;
  cin>>ch;
  cout<<static_cast<int>(ch)<<endl;
  return 0;
}
```

Question 10: ASCII to Character

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int x;
  cin>>x;
  cout<<char(x)<<endl;
  return 0;
}
```

Question 11: Lowercase to Uppercase

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  char ch;
  cin>>ch;
  cout<<char(ch-32)<<endl;
  return 0;
}
```

Question 12: Uppercase to Lowercase

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  char ch;
  cin>>ch;
  cout<<char(ch+32)<<endl;
  return 0;
}
```

Question 13: Digit Character to Integer

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  char ch;
  cin>>ch;
  cout<<ch-'0'<<endl;
  return 0;
}
```

Question 14: Integer to Character

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int n;
  cin>>n;
  cout<<char(n+'0')<<endl;
  return 0;
}
```

Question 15: Difference Between ++x and x++

```
#include <iostream>
#include <string>
```

```
using namespace std;
int main()
{
  int x=5;
  cout<<++x<<endl;
  cout<<x++<<endl;
  cout<<x<<endl;
  return 0;
}
```

Question 16: Variable Update Challenge

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b,c;
  cin>>a>>b>>c;
  a+=5;
  b-=2;
  c*=3;
  a/=2;
  cout<<a<<" "<<b<<" "<<c<<endl;
  return 0;
}
```

Question 17: Floating Point Division

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b;
  cin>>a>>b;
  cout<<a/b<<endl;
  cout<<static_cast<double>(a)/b<<endl;
  return 0;
}
```

Question 18: Full Name and Address

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  string first,last,address;
  cin>>first>>last;
  cin.ignore();
  getline(cin,address);
  cout<<first<<" "<<last<<endl;
  cout<<address<<endl;
  return 0;
```

```
}
```

Question 19: Mini Calculator

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  int a,b;
  cin>>a>>b;
  cout<<"Sum = "<<a+b<<endl;
  cout<<"Difference = "<<a-b<<endl;
  cout<<"Product = "<<a*b<<endl;
  cout<<"Division = "<<a/b<<endl;
  cout<<"Remainder = "<<a%b<<endl;
  return 0;
}
```

Question 20: Personal Information Card

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
  string name,college,branch;
  int age;
  double cgpa;
  getline(cin,name);
  cin>>age;
  cin.ignore();
  getline(cin,college);
  getline(cin,branch);
  cin>>cgpa;
  cout<<"========== Student Profile =========="<<endl;
  cout<<"Name : "<<name<<endl;
  cout<<"Age : "<<age<<endl;
  cout<<"College : "<<college<<endl;
  cout<<"Branch : "<<branch<<endl;
  cout<<"CGPA : "<<cgpa<<endl;
  cout<<"====================================="<<endl;
  return 0;
}
```

End of Day 3 Practice Set

NextGen Data Minds